

M002 Extension 4 — Worksheet (Beginner)

Topic: Add Two AND Conditions · **Course:** Maze Madness · **Level:** Extension (Beginner) · **Time:** about 35 minutes

This is a **challenge** for students who finished the weekly mazes. The agent already knows **four rules** for following redstone through a 3D maze. But it is **not finished** — at two spots it gets stuck because one redstone signal is not enough. You will look at each stuck spot, see **which two signals** are on, and add your own **AND** rule (two things true at once). Then get the agent to the goal.

1 · The Four Rules the Agent Knows

Each rule checks **one** redstone signal and does **one** thing.

```
if redstone on the LEFT:
    turn RIGHT
if redstone on the RIGHT:
    turn LEFT
if redstone BELOW:
    move UP
if redstone ABOVE:
    move DOWN
```

Draw a line to match each signal to its action. You will reuse these actions in your own rules.

Signal		Action
redstone on the left	→	move up
redstone below	→	turn right
redstone above	→	turn left
redstone on the right	→	move down

2 · AND means BOTH

AND is true only when **both** things are true.

```
if redstone on the LEFT AND redstone BELOW:  
    turn right  
    move up
```

(This is just an example.) There is redstone on the left, but NOT below. Is the AND true? Circle one: yes · no

There is redstone on the left AND redstone below. Is the AND true? Circle one: yes · no

When the AND is true, the agent does the action for *each* signal — so it does TWO things. How many actions does an AND rule with two signals need?

3 · Look at the Stuck Spots 🔍

Open the **M002 Complete the Maze** world. Type **run** . The agent stops at **two** spots because two signals are on at once. Walk to each spot and look around the agent.

Which signals are on at each stuck spot? Tick all you see.

	left	right	below	above
Stuck spot 1				
Stuck spot 2				

4 · Write Your Two AND Rules

Use the two signals you ticked for each spot, and the actions from the table in section 1.

Stuck spot 1 — fill in the two signals and the two actions:

if redstone on the _____ AND redstone _____:

Stuck spot 2 — write the whole rule yourself:

5 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version. Circle what is different and write one short sentence about the bug.

Pair A — A rule should fire only when **both** signals are on.

```
# clean
if redstone on the LEFT AND redstone BELOW:
    turn right
    move up
```

```
# buggy
if redstone on the LEFT OR redstone BELOW:
    turn right
    move up
```

What is wrong? With OR, how many of the two does the agent need?

Pair B — A rule with two signals needs **two** actions.

```
# clean
turn right
move up
```

```
# buggy
turn right
```

What is wrong? What does the agent forget to do?

6 · Finish the Whole Maze 📷

Type **run** and watch the agent. The goal is to reach the **very end**. If it gets stuck, fix your AND rules and run again.

When the agent finishes, come back here. Send a photo or video of the agent at the goal, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

The agent already knew four rules, like ...

At the first stuck spot the two signals were ... and ...

An AND rule means the agent needs ...

7 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while the agent solves the maze. Talk like you are teaching a friend. Try to use these words: **redstone, AND, both, turn, move up.**

1. Show the maze and the redstone.
2. Type **run** and show the agent following it.
3. Say out loud the two signals your AND rule needs.
4. Show the agent reaching the goal.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.